

LoRA vs. Full Fine-Tuning

A Theoretical Perspective on Generalization

Ali Zindari^{*1,2} Rotem Mulayoff^{*2} Sebastian U. Stich²

¹Universität des Saarlandes · ²CISPA Helmholtz Center for Information Security, Saarbrücken, Germany · * equal contribution

CONTACT {ali.zindari, rotem.mulayoff, stich}@cispa.de

TL;DR We derive **excess-risk bounds** for LoRA and Full FT in linear regression. In both the over- and under-determined regimes, LoRA reduces variance from $\Theta(d_x d_y/n)$ to $\mathcal{O}(r \max\{d_x, d_y\}/n)$. When the task shift Δ^* is effectively **low-rank**, LoRA **strictly outperforms** FFT; the gap is largest in *high-noise / few-sample* regimes. Qwen2.5 fine-tuning experiments confirm the linear theory.

1 Problem setup

Linear regression with n i.i.d. samples $\mathcal{D} = \{(\mathbf{x}_i, \mathbf{y}_i)\}_{i=1}^n$:

$$\mathbf{y}_i = \mathbf{A}^* \mathbf{x}_i + \boldsymbol{\varepsilon}_i, \quad \boldsymbol{\varepsilon}_i \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}_{\varepsilon\varepsilon}), \quad \Delta^* := \mathbf{A}^* - \mathbf{A}_0.$$

Stacking samples as columns gives the data matrices:

$$\mathbf{X} = [\mathbf{x}_1, \dots, \mathbf{x}_n] \in \mathbb{R}^{d_x \times n}, \quad \mathbf{Y} = [\mathbf{y}_1, \dots, \mathbf{y}_n] \in \mathbb{R}^{d_y \times n}.$$

Both methods solve $\min_{\mathbf{A}} \|\mathbf{A}\mathbf{X} - \mathbf{Y}\|_{\text{F}}^2$. They differ only in the constraint on $\mathbf{A}$, and we always pick the *min-norm* solution.

FFT (min-norm)

$$\hat{\mathbf{A}}_{\text{FFT}} = \mathbf{A}_0 + \widehat{\Delta}_{\text{FFT}}, \quad \widehat{\Delta}_{\text{FFT}} := (\mathbf{Y} - \mathbf{A}_0 \mathbf{X}) \mathbf{X}^\dagger.$$

LoRA (rank- r , Sondermann)

$$\hat{\mathbf{A}}_{\text{LoRA}} = \mathbf{A}_0 + \text{trunc}_r \left\{ \widehat{\Delta}_{\text{FFT}} \widehat{\boldsymbol{\Sigma}}_{\text{xx}}^{1/2} \right\} \left(\widehat{\boldsymbol{\Sigma}}_{\text{xx}}^{1/2} \right)^\dagger.$$

LoRA's update is the FFT update *projected onto its top- r singular components* in the data geometry, then mapped back. Both estimators have closed form; we study their *expected excess risk*.

Lemma 5 (Excess-risk identity)

$$\mathcal{R}(\hat{\mathbf{A}}) - \mathcal{R}(\mathbf{A}^*) = \|(\hat{\mathbf{A}} - \mathbf{A}^*) \boldsymbol{\Sigma}_{\text{xx}}^{1/2}\|_{\text{F}}^2.$$

2 Overdetermined regime $n \geq d_x$

Theorem 1 (FFT excess risk)

$$\begin{aligned} \mathbb{E}[\mathcal{R}(\hat{\mathbf{A}}_{\text{FFT}})] - \mathcal{R}(\mathbf{A}^*) &= \frac{1}{n} \text{Tr}(\boldsymbol{\Sigma}_{\varepsilon\varepsilon}) \mathbb{E}[\text{Tr}(\boldsymbol{\Sigma}_{\text{xx}} \widehat{\boldsymbol{\Sigma}}_{\text{xx}}^{-1})] \\ &\geq \bar{\sigma}_\varepsilon^2 \frac{d_x d_y}{n}, \quad \bar{\sigma}_\varepsilon^2 := \frac{1}{d_y} \mathbb{E}[\|\boldsymbol{\varepsilon}\|^2]. \end{aligned}$$

When $n \geq d_x$, FFT coincides with OLS: it does not depend on $\mathbf{A}_0$ and is *unbiased*. The bound scales as $\Omega(d_x d_y/n)$ and blows up at the interpolation threshold $n \rightarrow d_x$.

Proposition 3 (Gaussian case, FFT)

If $\mathbf{x}_i \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}_{\text{xx}})$ i.i.d. and $\boldsymbol{\Sigma}_{\varepsilon\varepsilon} \succeq \mathbf{0}$:

$$\mathbb{E}[\mathcal{R}(\hat{\mathbf{A}}_{\text{FFT}})] - \mathcal{R}(\mathbf{A}^*) = \text{Tr}(\boldsymbol{\Sigma}_{\varepsilon\varepsilon}) \frac{d_x}{n - d_x - 1}.$$

Theorem 2 (LoRA excess risk)

$$\begin{aligned} \mathbb{E}[\mathcal{R}(\hat{\mathbf{A}}_{\text{LoRA}})] - \mathcal{R}(\mathbf{A}^*) &\lesssim \frac{r \max\{d_x, d_y\}}{n} \lambda_{\max}(\boldsymbol{\Sigma}_{\varepsilon\varepsilon}) \mathbb{E}[\lambda_{\max}(\boldsymbol{\Sigma}_{\text{xx}} \widehat{\boldsymbol{\Sigma}}_{\text{xx}}^{-1})] \\ &+ \lambda_{\max}(\boldsymbol{\Sigma}_{\text{xx}}) \left(r \sigma_{r+1}^2(\Delta^*) + \sum_{i>r} \sigma_i^2(\Delta^*) \right) \\ &+ r \sigma_{r+1}^2(\Delta^*) \mathbb{E}[\lambda_{\max}(\widehat{\boldsymbol{\Sigma}}_{\text{xx}}) \lambda_{\max}(\boldsymbol{\Sigma}_{\text{xx}} \widehat{\boldsymbol{\Sigma}}_{\text{xx}}^{-1})]. \end{aligned}$$

The first term is LoRA's *variance*. The remaining two terms are the *bias* incurred by the rank- r truncation: this error is *inevitable*, since any rank- r adaptation must misfit the spectral tail of Δ^* (terms involving $\sigma_{r+1}(\Delta^*)$ and $\sum_{i>r} \sigma_i^2(\Delta^*)$). The bias is small exactly when Δ^* has rapidly decaying singular values.

Proposition 4 (asymptotic LoRA variance)

Under the Gaussian setting of Prop. 3, the variance term in Thm. 2 is asymptotically equivalent to

$$r \max\{d_x, d_y\} (\sqrt{n} - \sqrt{d_x})^{-2} \lambda_{\max}(\boldsymbol{\Sigma}_{\varepsilon\varepsilon}).$$

Far from the interpolation threshold ($n \gg d_x$), with bounded $\lambda_{\max}(\boldsymbol{\Sigma}_{\varepsilon\varepsilon})$ but non-vanishing $\bar{\sigma}_\varepsilon^2$ (e.g. isotropic noise), this is $\mathcal{O}(r \max\{d_x, d_y\}/n)$, versus FFT's $\Omega(d_x d_y/n)$. Since r is typically much smaller than $\min\{d_x, d_y\}$, LoRA achieves a *substantial reduction in variance*.

Corollary 1 (low-rank shift)

If $\text{rank}(\Delta^*) \leq r$:

$$\mathbb{E}[\mathcal{R}(\hat{\mathbf{A}}_{\text{LoRA}})] - \mathcal{R}(\mathbf{A}^*) \lesssim \frac{r \max\{d_x, d_y\}}{n} \lambda_{\max}(\boldsymbol{\Sigma}_{\varepsilon\varepsilon}) \mathbb{E}[\lambda_{\max}(\boldsymbol{\Sigma}_{\text{xx}} \widehat{\boldsymbol{\Sigma}}_{\text{xx}}^{-1})].$$

Bias-variance tradeoff. As r grows, the truncation bias decreases but the variance grows linearly in r . When the spectrum of Δ^* decays fast, even a small r pays a tiny bias and reaps the full variance reduction. When the spectrum is flat, no $r < \min\{d_x, d_y\}$ can compete with FFT. In modern LLMs the spectrum is heavily concentrated in a few leading directions, making the low-rank assumption mild.

FFT vs. LoRA. $\Theta(d_x d_y/n)$ vs. $\mathcal{O}(r \max\{d_x, d_y\}/n)$: when $r \ll \min\{d_x, d_y\}$, LoRA achieves a *substantial variance reduction*.

3 Underdetermined regime $n < d_x$

Theorem 3 (FFT excess risk)

$$\mathbb{E}[\mathcal{R}(\hat{\mathbf{A}}_{\text{FFT}})] - \mathcal{R}(\mathbf{A}^*) = \underbrace{\mathbb{E}[\|\Delta^* \mathbf{P}_{\mathbf{X}}^\perp \boldsymbol{\Sigma}_{\text{xx}}^{1/2}\|_{\text{F}}^2]}_{\text{FFT Bias}} + \underbrace{\frac{1}{n} \text{Tr}(\boldsymbol{\Sigma}_{\varepsilon\varepsilon}) \mathbb{E}[\text{Tr}(\boldsymbol{\Sigma}_{\text{xx}} \widehat{\boldsymbol{\Sigma}}_{\text{xx}}^{-1})]}_{\text{FFT Variance}}.$$

Prop. 5 (Gaussian case, FFT)

$$\text{Bias} = \|\Delta^*\|_{\text{F}}^2 \frac{d_x - n}{d_x}, \quad \text{Variance} = \text{Tr}(\boldsymbol{\Sigma}_{\varepsilon\varepsilon}) \frac{n}{d_x - n - 1}.$$

FFT is now *biased*: the data only constrain $\mathbf{A}$ in the row space of $\mathbf{X}$, and the rest of Δ^* leaks into the risk. Bias decreases linearly with n , while variance grows with n ; they balance near $n \rightarrow d_x$.

Theorem 4 (LoRA excess risk)

$$\begin{aligned} \mathbb{E}[\mathcal{R}(\hat{\mathbf{A}}_{\text{LoRA}})] - \mathcal{R}(\mathbf{A}^*) &\lesssim \mathbb{E}[\|\Delta^* \mathbf{P}_{\mathbf{X}}^\perp \boldsymbol{\Sigma}_{\text{xx}}^{1/2}\|_{\text{F}}^2] \\ &+ \frac{r \max\{n, d_y\}}{n} \lambda_{\max}(\boldsymbol{\Sigma}_{\varepsilon\varepsilon}) \mathbb{E}[\lambda_{\max}(\boldsymbol{\Sigma}_{\text{xx}} \widehat{\boldsymbol{\Sigma}}_{\text{xx}}^{-1})] \\ &+ \lambda_{\max}(\boldsymbol{\Sigma}_{\text{xx}}) \left(r \sigma_{r+1}^2(\Delta^*) + \sum_{i>r}^{\min\{n, d_y\}} \sigma_i^2(\Delta^*) \right) \\ &+ r \sigma_{r+1}^2(\Delta^*) \mathbb{E}[\lambda_{\max}(\widehat{\boldsymbol{\Sigma}}_{\text{xx}}) \lambda_{\max}(\boldsymbol{\Sigma}_{\text{xx}} \widehat{\boldsymbol{\Sigma}}_{\text{xx}}^{-1})]. \end{aligned}$$

The first term is the *same* bias as FFT (Thm. 3). The last two are truncation error from the rank- r constraint, which vanish if $\text{rank}(\Delta^*) \leq r$.

Proposition 6 (asymptotic LoRA variance)

Assume $d_x > n + 1$, $\mathbf{x}_i \sim \mathcal{N}(\mathbf{0}, \sigma_x^2 \mathbf{I})$ i.i.d., and $\boldsymbol{\Sigma}_{\varepsilon\varepsilon} \succeq \mathbf{0}$. The variance term in Thm. 4 is asymptotically equivalent to

$$r \max\{n, d_y\} (\sqrt{d_x} - \sqrt{n})^{-2} \lambda_{\max}(\boldsymbol{\Sigma}_{\varepsilon\varepsilon}).$$

Corollary 2 (low-rank shift)

If $\text{rank}(\Delta^*) \leq r < n$:

$$\mathbb{E}[\mathcal{R}(\hat{\mathbf{A}}_{\text{LoRA}})] - \mathcal{R}(\mathbf{A}^*) \lesssim \mathbb{E}[\|\Delta^* \mathbf{P}_{\mathbf{X}}^\perp \boldsymbol{\Sigma}_{\text{xx}}^{1/2}\|_{\text{F}}^2] + \frac{r \max\{n, d_y\}}{n} \lambda_{\max}(\boldsymbol{\Sigma}_{\varepsilon\varepsilon}) \mathbb{E}[\lambda_{\max}(\boldsymbol{\Sigma}_{\text{xx}} \widehat{\boldsymbol{\Sigma}}_{\text{xx}}^{-1})].$$

Bias parity, variance gain. LoRA inherits FFT's bias exactly, but enjoys the $r / \min\{d_x, d_y\}$ variance reduction. When Δ^* 's spectrum decays fast, LoRA Pareto-dominates FFT in both regimes.

4 Linear regression experiment

Isotropic Gaussian features; $\Delta^* = \frac{1}{\sqrt{r^*}} \mathbf{U}\mathbf{V}$ with Gaussian $\mathbf{U}, \mathbf{V}$. 100 seeds; shaded ± 1 s.d.

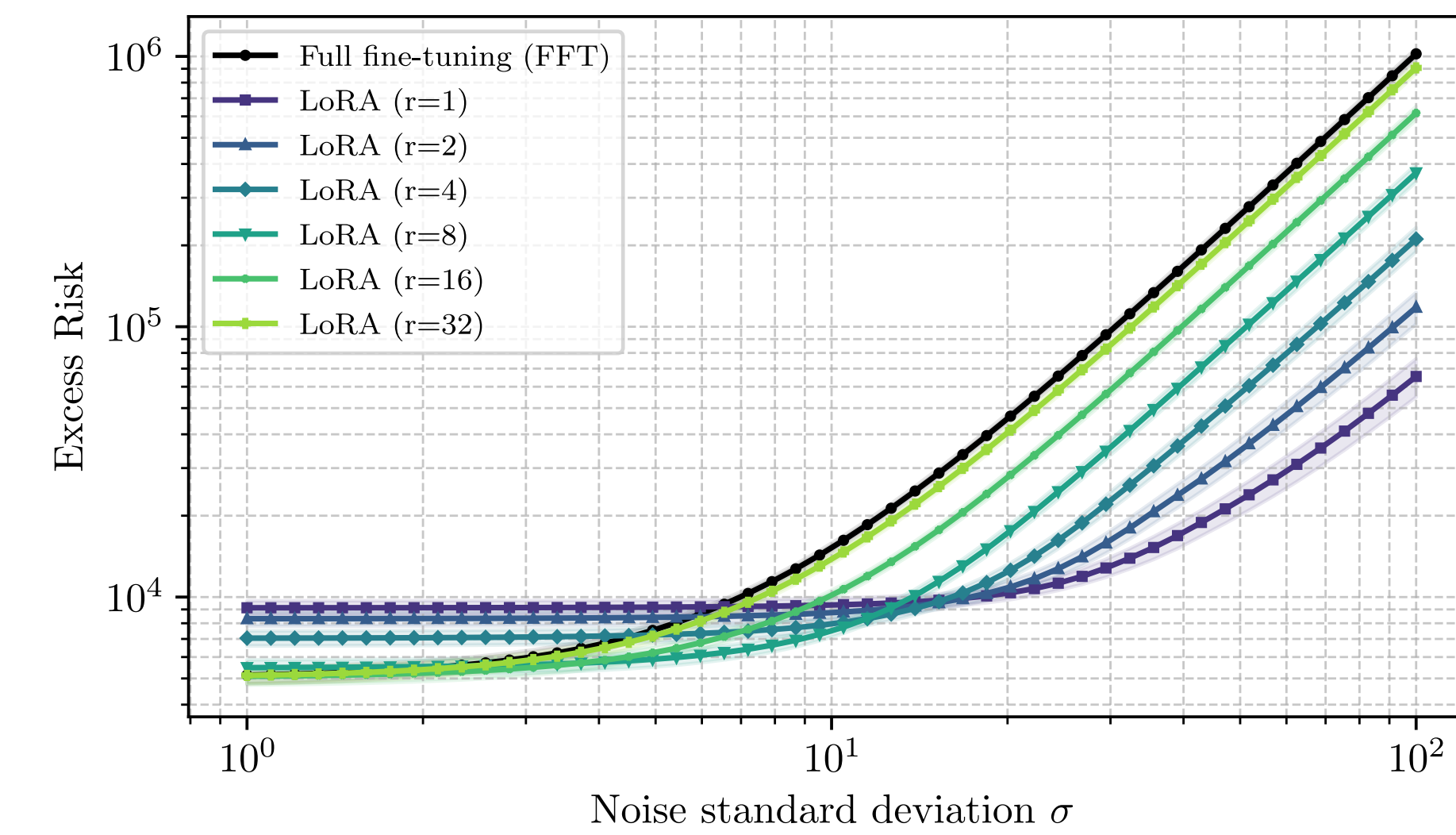
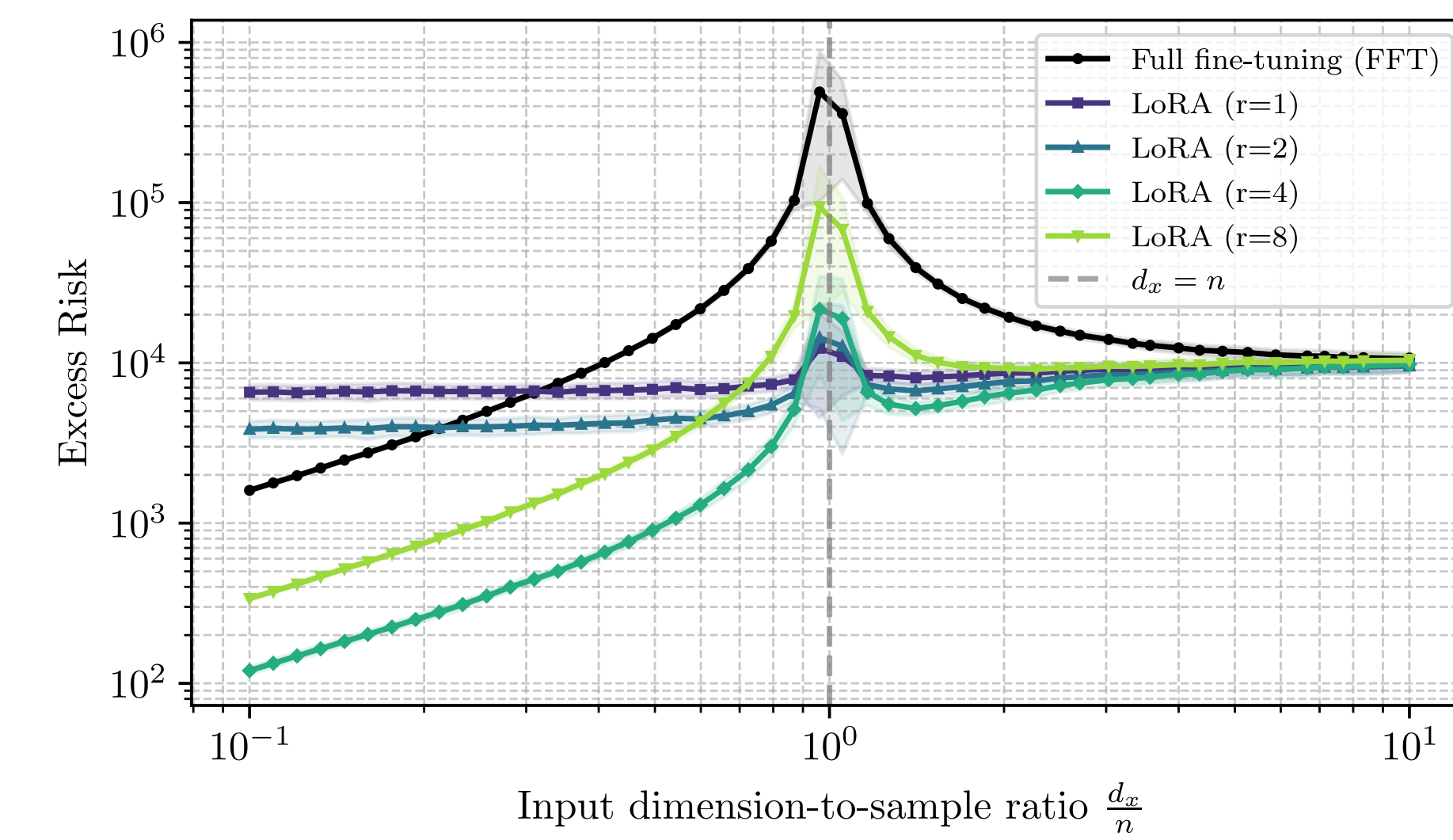


Fig. 1. Top: excess risk vs. d_x/n with $r^* = 4$. Bottom: excess risk vs. noise $\sigma \in [1, 100]$, $d_x = d_y = 100$, $n = 50$, $r^* = 10$.

- $r \geq r^*$: LoRA beats FFT at every d_x/n ; the optimum is $r = r^*$.
- $r < r^*$: rank-bias dominates and LoRA can hurt; risk is nearly flat in d_x/n .
- High noise**: FFT explodes with σ ; low-rank LoRA stays bounded.
- Interpolation peak** at $d_x/n = 1$ matches the $\sqrt{n} - \sqrt{d_x}$ scaling of Props. 4 and 6.

5 LLM experiment: label noise

Qwen2.5-0.5B fine-tuned with various ranks on *BoolQ* & *CommonsenseQA*; 3 seeds. The noise level controls the variance term in our excess-risk bounds: larger $\text{Tr}(\boldsymbol{\Sigma}_{\varepsilon\varepsilon})$ means the variance gap between FFT and LoRA dominates.

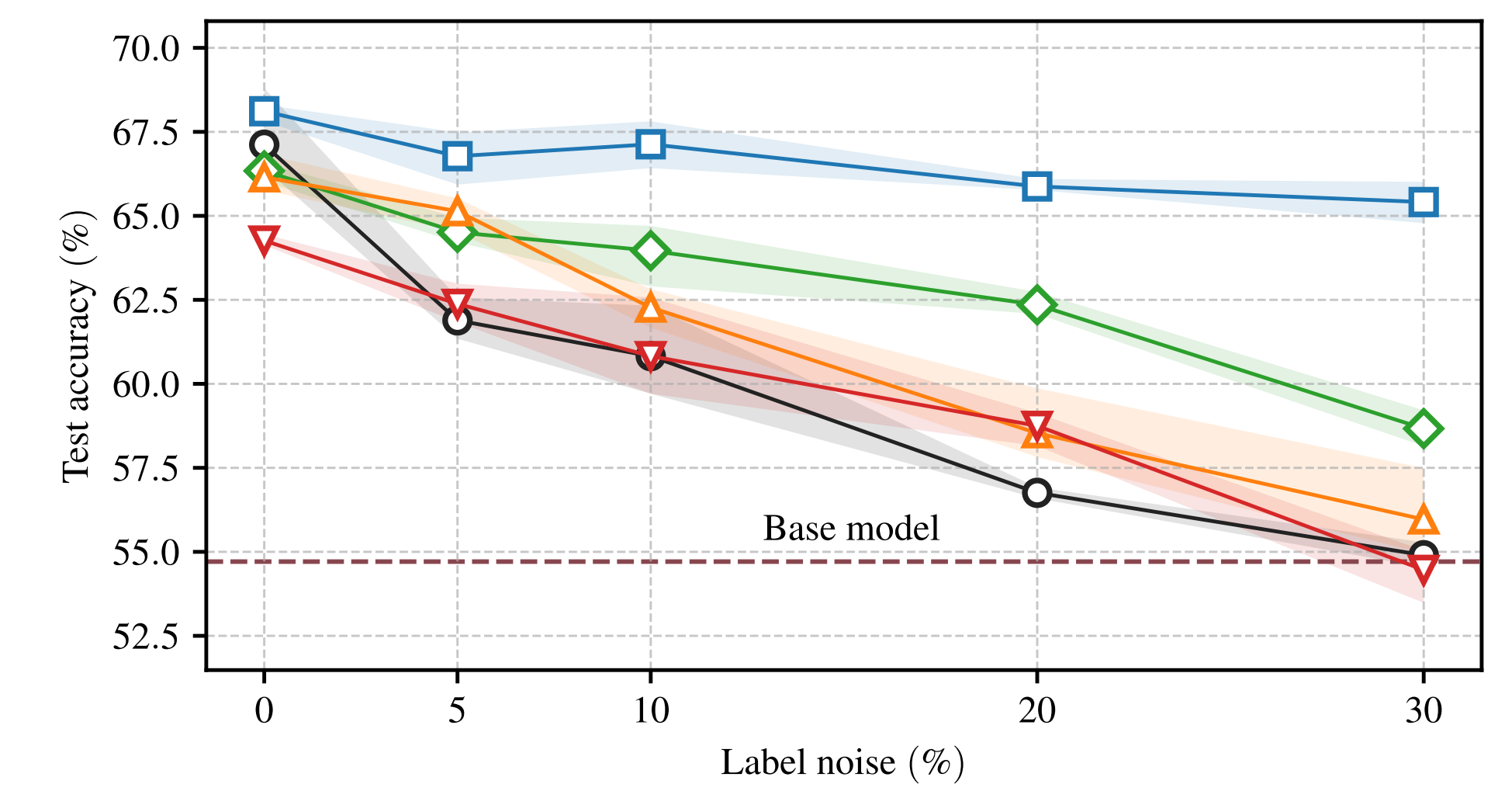
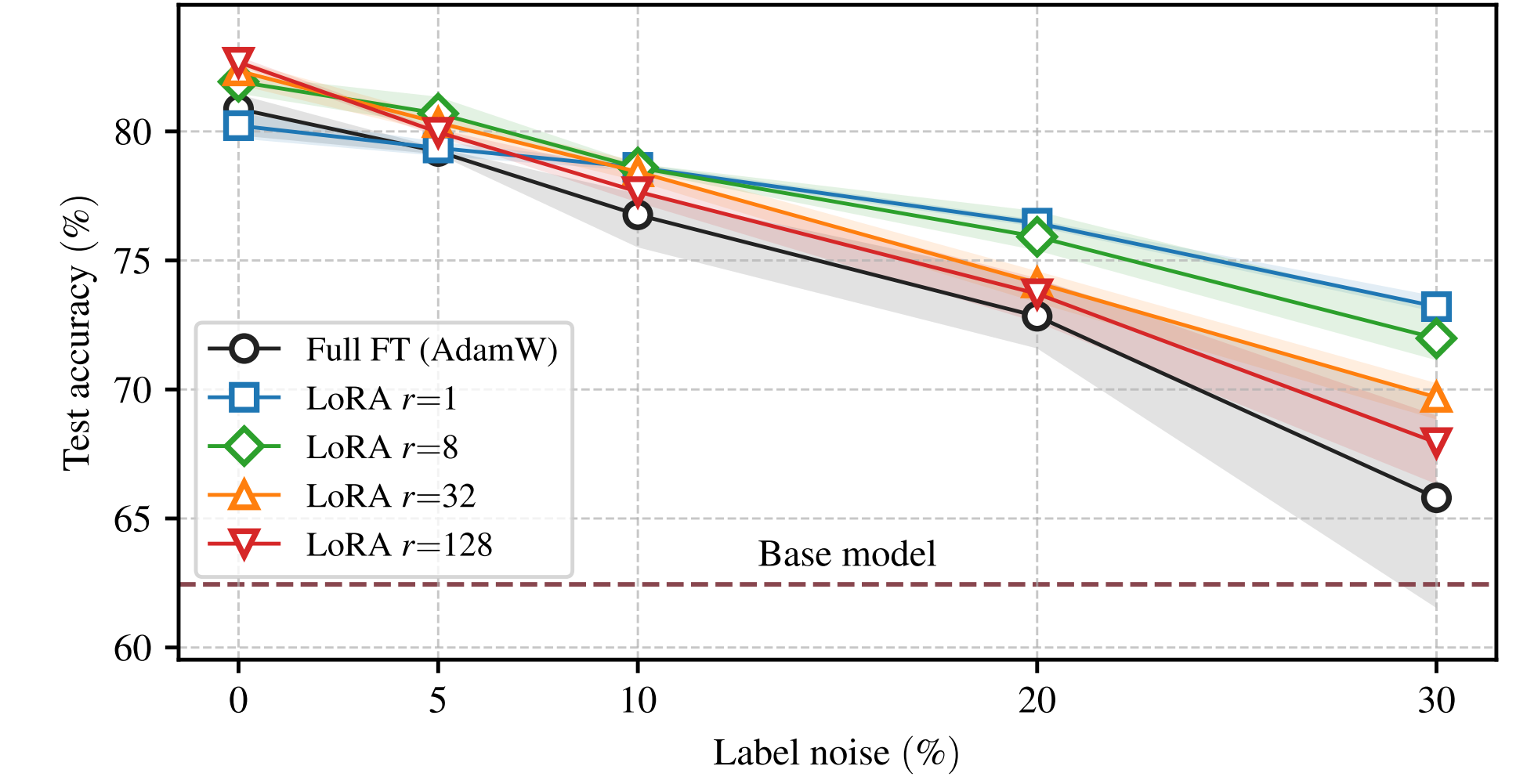


Fig. 2. Test accuracy vs. label noise on *BoolQ* (top) and *CommonsenseQA* (bottom). Strong-noise regime: LoRA outperforms Full FT (Cor. 1).

- Weak noise.** No consistent rank ordering: a moderate r is optimal, the classical bias/variance tradeoff.
- Strong noise.** Test accuracy decreases *monotonically* with r . Restricting the number of trainable directions improves generalization.
- Why?** Variance dominates the excess risk and LoRA scales it down by $r / \min\{d_x, d_y\}$ exactly as Cor. 1 predicts.

Two regimes (theory). When $\text{Tr}(\boldsymbol{\Sigma}_{\varepsilon\varepsilon}) \gg \|\Delta^*\|_{\text{F}}^2$ the variance term dominates the bias and adding samples can hurt. When $\text{Tr}(\boldsymbol{\Sigma}_{\varepsilon\varepsilon}) \ll \|\Delta^*\|_{\text{F}}^2$ the bias governs the risk. The plots above span both regimes.

6 LLM experiment: sample size & spectrum

Qwen2.5-1.5B with various LoRA ranks across training-set sizes. Lower panels: normalized cumulative spectral energy of the estimated Δ^* .

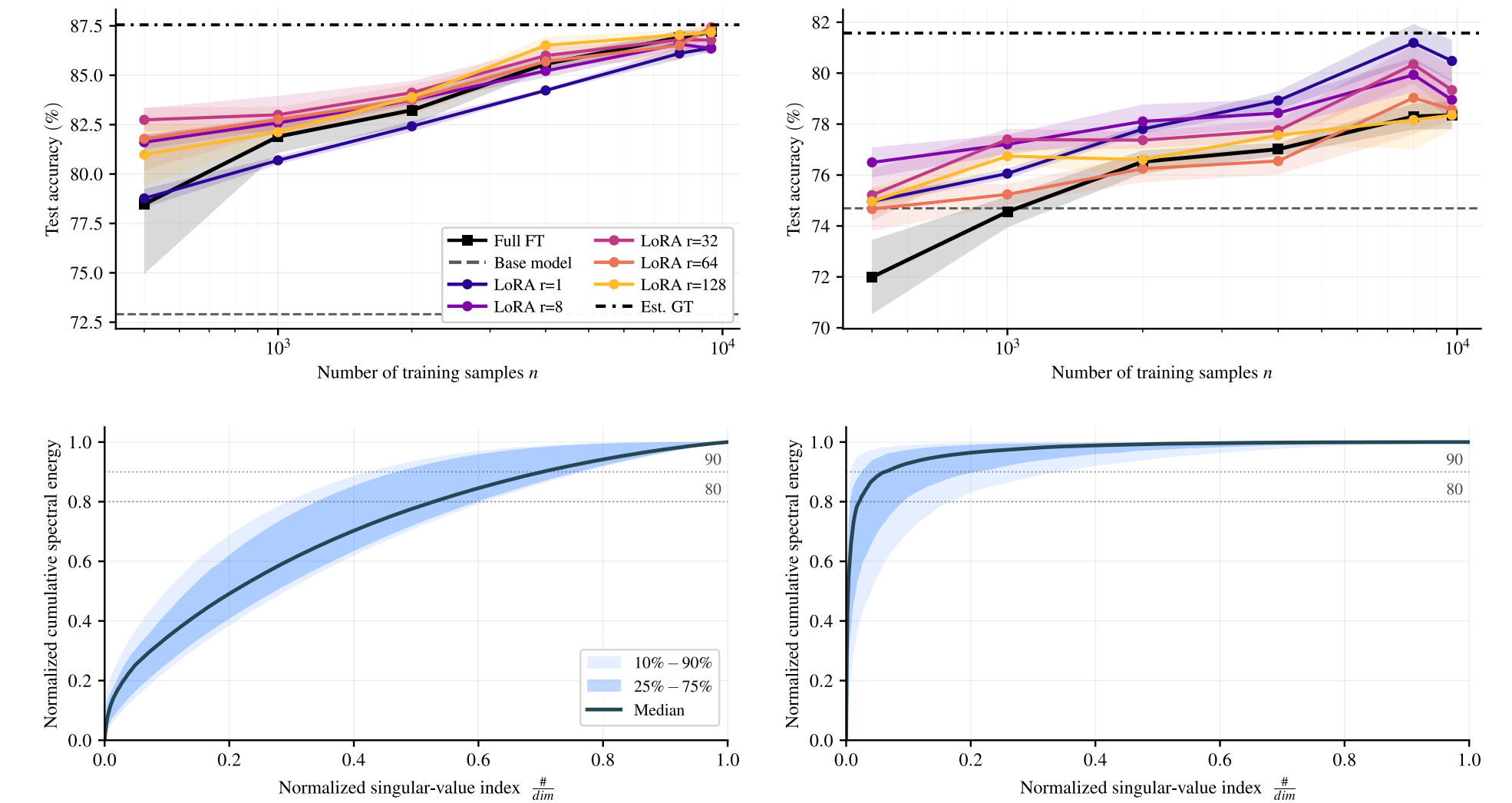


Fig. 3. Top: accuracy vs. n . Bottom: cumulative spectral energy of Δ^* . *BoolQ* (left): heavier tail. *CommonsenseQA* (right): rapid decay $\Rightarrow \Delta^*$ effectively low-rank.

- CommonsenseQA.** Rapid spectral decay, so Δ^* is effectively low-rank, and LoRA *significantly* outperforms FFT.
- BoolQ.** Heavy spectral tail, so Δ^* is poorly approximated by rank r , and LoRA matches FFT but does not beat it.
- Spectral decay = practical knob.** The cumulative-energy curve of an estimated Δ^* is a cheap diagnostic for when LoRA will help.
- Confirmation across model sizes.** The qualitative trends in Figs. 2 and 3 reproduce on Qwen2.5-0.5B / 1.5B / 3B (App. VI), supporting the linear theory.

Takeaways

- First **excess-risk analysis** of LoRA covering both over- and under-determined regimes.
- FFT $\Theta(d_x d_y/n)$ vs. LoRA $\mathcal{O}(r \max\{d_x, d_y\}/n)$: variance reduced by $r / \min\{d_x, d_y\}$.
- LoRA **strictly outperforms FFT** when Δ^* is effectively low-rank; gap grows in high-noise / few-sample regimes.
- Qwen2.5 experiments match the linear theory: label noise & spectral decay decide the winner.